

The defect-free boron arsenide has a record-high thermal conductivity, more than three times faster at conducting heat than currently used materials, such as silicon-carbide and copper, so that heat that would otherwise concentrate in hot-spots is quickly flushed away.

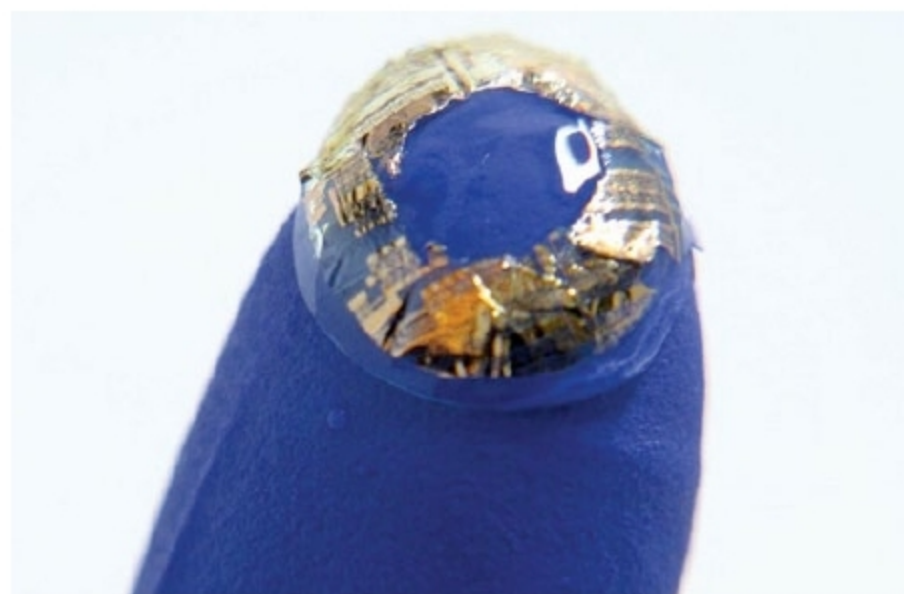
Yongjie Hu, assistant professor of mechanical and aerospace engineering at UCLA, says, "This material could improve performance and reduce energy demand in all kinds of electronics, from small devices to the most advanced computer data centre equipment. It can be integrated into current manufacturing processes because of its semiconductor properties and the demonstrated capability to scale this technology. It could replace current state-of-the-art semiconductor materials for computers and revolutionise the electronics industry."

Soft contact lenses for monitoring glucose, medical conditions, delivering medications

Researchers from Purdue University have developed soft contact lenses that not only correct vision, but also monitor glucose and medical conditions. These be used for ocular pain relief or drug delivery.

Until recently, sensors or other technology could not be used for soft contact lenses, because the technology required a rigid, planar surface incompatible with the soft, curved shape of a contact lens. The team, led by Chi Hwan Lee, assistant professor of biomedical engineering and mechanical engineering, created a novel method for attaching sensors and other small devices to soft contact lenses.

Lee says, "We developed a unique technology that enables the integration of thin-film sensors with commercially-available soft contact lenses. These current hydrogel-based contacts serve as the perfect platform for smart lens systems due to their high degree of comfortability, bio-compatibility, breathability and long-term wearability. Before our discovery, it was challenging to fabricate high-performance electronics on commercially-available soft contact lenses."



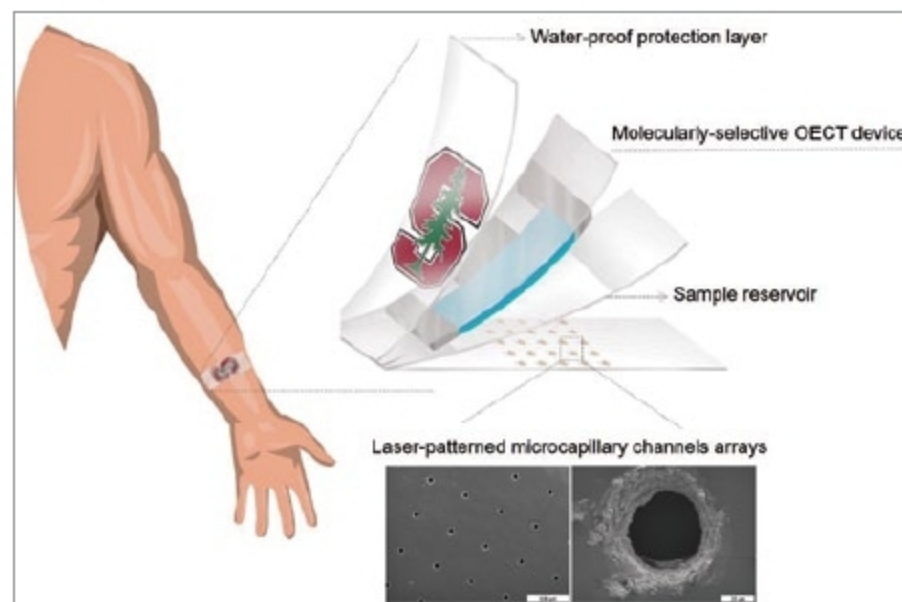
The technology will form a basis to further extend the functionality of smart soft contact lenses system for many other envisioned applications, including controlled release of ocular drugs, eye-wearable night vision and augmented reality (Credit: Purdue University)

Sensors embedded on the soft contact lenses detect levels of glucose, lactate and pH value in a continuous manner, providing information associated with diabetes, hypoxia and underlying ocular tissue health. With the ability to combine soft, silicon-based contact lenses with a variety of different semiconductor devices, numerous advanced eye care applications are now possible.

Wearable device that can detect stress levels through sweat

Hormone cortisol rises and falls naturally throughout the day and can spike in response to stress. Current methods for measuring cortisol levels require waiting several days for results from a lab.

Now, a group of scientists led by Alberto Salleo, a materials scientist at Stanford University, has created a stretchy patch that, when applied directly to the skin, wicks up sweat and assesses how much cortisol a person is producing.



Drawing showing details of the layers contained in the cortisol biosensor and two close-up images of the holes in the bottom of the sensor that wick in sweat (Credit: Onur Parlak, Stanford University)

Onur Parlak, a post-doctoral scholar in Salleo Lab (part of Salleo Research Group, Stanford University) and lead author of the paper, says, "We are interested in sweat sensing, because it offers non-invasive and continuous monitoring of various biomarkers for a range of physiological conditions. This offers a novel approach for the early detection of various diseases and evaluation of sports performance."

Parlak built this stretchable, rectangular sensor around a membrane that specifically binds only to cortisol. Stuck to the skin, it sucks in sweat passively through holes in the bottom of the patch. Sweat pools in a reservoir, which is covered by the cortisol-sensitive membrane. Charged ions like sodium or potassium—also found in sweat—pass through the membrane unless these are blocked by cortisol. It is those backed up charged ions the sensor detects, not the cortisol itself. To top it all, this is a waterproof layer that protects the patch from contamination.

All a user needs to see cortisol levels is to sweat (enough to glisten), apply the patch and connect it to a device for analysis, which gives results in seconds.